TECH2100 — Introduction to Information Networks

Assessment 1

Report Evaluation

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30 March 2026

Contents

1 Potential reputation, financial and legal loss of current network architecture.

Along the years, technologies have been experiencing a rapidly evolution and it close to what is known as Moore's law. Intel co-founder Gordon Moore predicted a doubling of transistors every year for the next 10 years in his original paper published in 1965. This boom in the evolution is followed by its demand adding extra stress to the network infrastructure and a major outage in entire Australia could potentially cause a damage of AUD $29,828,489 per hour (Figure 01).

Figure 01: Data from the Netblocks, the economic impact of a single hypothetical internet disruption of the specified type, location and duration.

2 Section Two

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas (Stallings, 2022).

2.1 Subsection A

Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae. Cras sit amet nibh libero, in gravida nulla (Forouzan, 2022). Nulla vel metus scelerisque ante sollicitudin.

2.2 Subsection B — Figure Example

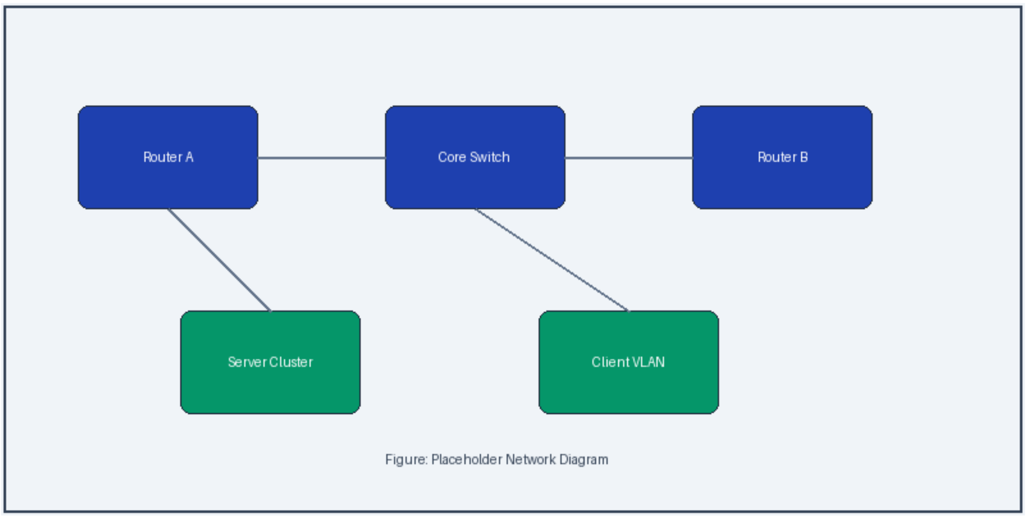

Figure 1: Placeholder diagram showing a sample layout (adapted from Kurose and Ross 2021).

As shown in Figure 1, the layout separates components into distinct layers. Tanenbaum et al. (2021) recommend this approach for medium-scale deployments.

2.3 Subsection C

Fusce dapibus, tellus ac cursus commodo, tortor mauris condimentum nibh, ut fermentum massa justo sit amet risus (Kreutz et al., 2015).

3 Section Three

3.1 Subsection A

Maecenas sed diam eget risus varius blandit sit amet non magna. Integer posuere erat a ante venenatis dapibus posuere velit aliquet (Zhang et al., 2019). Praesent commodo cursus magna, vel scelerisque nisl consectetur et.

3.2 Subsection B

Cras mattis consectetur purus sit amet fermentum. Donec sed odio dui. Aenean lacinia bibendum nulla sed consectetur (Boutaba et al., 2018).

3.3 Subsection C

Nullam quis risus eget urna mollis ornare vel eu leo. Morbi leo risus, porta ac consectetur ac, vestibulum at eros (Cisco Systems, 2023).

4 Section Four

4.1 Subsection A

Etiam porta sem malesuada magna mollis euismod. Vivamus sagittis lacus vel augue laoreet rutrum faucibus dolor auctor (Cisco Systems, 2020).

4.2 Subsection B

Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus (IEEE Standards Association, 2022). Donec ullamcorper nulla non metus auctor fringilla.

4.3 Subsection C

Sed posuere consectetur est at lobortis. Cras justo odio, dapibus ut facilisis in, egestas eget quam (Akyildiz et al., 2014).

5 Section Five

5.1 Subsection A

Maecenas faucibus mollis interdum. Duis mollis, est non commodo luctus, nisi erat porttitor ligula, eget lacinia odio sem nec elit (Floridi et al., 2018). Nullam id dolor id nibh ultricies vehicula ut id elit.

5.2 Subsection B

Aenean eu leo quam. Pellentesque ornare sem lacinia quam venenatis vestibulum. Donec id elit non mi porta gravida at eget metus (National Institute of Standards and Technology, 2020).

5.3 Subsection C

Vivamus sagittis lacus vel augue laoreet rutrum faucibus dolor auctor. Cras mattis consectetur purus sit amet fermentum (Boutaba et al., 2018).

6 Conclusion

Donec sed odio dui. Etiam porta sem malesuada magna mollis euismod. Nullam quis risus eget urna mollis ornare vel eu leo. Integer posuere erat a ante venenatis dapibus posuere velit aliquet.

Reference List

Akyildiz, I. F., A. Lee, P. Wang, M. Luo & W. Chou. 'A roadmap for traffic engineering in SDN-OpenFlow networks'. *Computer Networks*, vol. 71, pp. 1–30, 2014.

Boutaba, R., M. A. Salahuddin, N. Limam, S. Ayoubi, N. Shahriar, F. Estrada-Solano & O. M. Caicedo. 'A comprehensive survey on machine learning for networking: evolution, applications and research opportunities'. *Journal of Internet Services and Applications*, vol. 9, no. 1, pp. 1–99, 2018.

Cisco Systems. 'Cisco annual internet report (2018–2023)'. *Cisco White Paper*, 2020. <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>. viewed 15 March 2026.

Cisco Systems. 'Quality of service design overview', 2023. <https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/QoS/QoSDesign.html>. viewed 14 March 2026.

Floridi, L., J. Cowls, M. Beltrametti, R. Chatila, P. Chazerand, V. Dignum, C. Luetge, R. Madelin, U. Pagallo, F. Rossi, B. Schafer, P. Valcke & E. Vayena. 'An ethical framework for a good AI society: opportunities, risks, principles, and recommendations'. *Minds and Machines*, vol. 28, no. 4, pp. 689–707, 2018.

Forouzan, B. A. *Data communications and networking*. McGraw-Hill Education, New York, 5th edn, 2022.

IEEE Standards Association. 'IEEE 802.3-2022 — Ethernet standard', 2022. <https://standards.ieee.org/standard/802_3-2022.html>. viewed 12 March 2026.

Kreutz, D., F. M. V. Ramos, P. E. Verissimo, C. E. Rothenberg, S. Azodolmolky & S. Uhlig. 'Software-defined networking: a comprehensive survey'. *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14–76, 2015.

Kurose, J. F. & K. W. Ross. *Computer networking: a top-down approach*. Pearson, Hoboken, NJ, 8th edn, 2021.

National Institute of Standards and Technology. 'Zero trust architecture'. In *NIST Special Publication 800-207*, Gaithersburg, MD, 2020. U.S. Department of Commerce. <https://doi.org/10.6028/NIST.SP.800-207>. viewed 10 March 2026.

Stallings, W. *Data and computer communications*. Pearson, Upper Saddle River, NJ, 10th edn, 2022.

Tanenbaum, A. S., N. Feamster & D. J. Wetherall. *Computer networks*. Pearson, Hoboken, NJ, 6th edn, 2021.

Zhang, C., P. Patras & H. Haddadi. 'Deep learning in mobile and wireless networking: a survey'. *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2224–2287, 2019.

A Appendix A: Diagrams

[Include full-size diagrams here if needed.]

B Appendix B: Supporting Material

[Include supporting material here.]